

Bio-inspired Embodied Language Processing

• Language

- universal in human society and unique to humans
- is compositional
- ⇒ Universality of language is a consequence of the fact that the human brain has evolved language processing systems

• Psychological / neurobiological models

- First systematic studies: relationship between brain and language
- Phrenology / Craniology:
 - Located language in anterior parts of the brain at the protrusion of the eye socket below the eye
 - Bumps of the skull taken to reflect areas of enlargement in the brain
- Empiricism: Broca: anatomical inspection of the brain

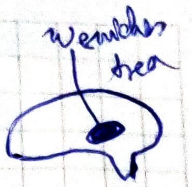
• Broca's Area: Motor & Production

- Localization of functions in the cerebral cortex
- Broca's aphasia: (Motor & Production)
 - Impaired language production
 - Sparse speech, nonfluent
 - Deficits of intonation and stress patterns
 - Lack of grammatical structure
 - Omitting function words



Broca's Area

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• Wernicke's Area

- Carl Wernicke: (Auditory & Comprehension)
- Area of the left superior temporal lobe
- Superior temporal area: where auditory words are stored
- Connection between Wernicke and Broca
- Wernicke's aphasia: impaired language comprehension

• Organisation of language in the brain

- Frontal lobe: contains Broca's area
 - ↳ Damage result in impaired language production
- Temporal lobe: contains Wernicke's area
 - ↳ Damage result in ~~impaired language production~~ impaired language comprehension

• Human language processing: Functions

- Comprehension: Maps from "sound to meaning" Dorsal

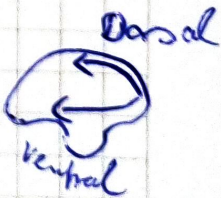
↳ Speech to Words ↳ Structures to meanings

- Production: Maps from "meaning to speech" Ventral

↳ Meaning to grammatical encoding

↳ Phonological encoding

↳ Articulation



- early models rejected: anatomic & linguistic underspecification

(↳ Imaging tools and new methods led to a major paradigm shift in models for language processing!) unmistak

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• Language in the brain

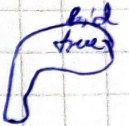
→ Language processing:

- Several pathways
- Re-using parts of the brain both in production and comprehension
- Spatial and temporal hierarchical abstraction

→ Language representation:

- Distributed semantic maps
- Embodied in multi-modal perception
- Involves whole brain

→ eg: Dual stream hypothesis for processing, semantic maps



• Models for language processing: Multi-layer associator

→ Computational model for auditory word recognition and motor control of speech ~~and~~ associator

↳ Multi-layer NN

↳ Mimics neuroanatomical connectivity and neurophysiological properties of language related areas

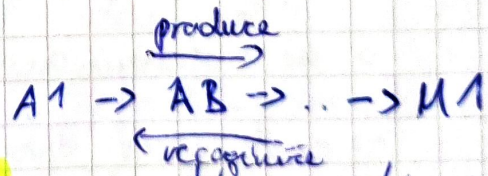
• Establishing associations

→ Neural Net for word learning

↳ A1: morpheme / auditory form

↳ M1: articulatory form

→ Sparseness: 25x25 cells each layer



via action-perception correlation

Hebbian rule: $\Delta w_{ij} = \alpha (x_i) (x_j)$

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• Language representation: Embodied model

→ Language is embodied and distributed over several areas of the brain

⇒ word - nets

→ Different areas active during processing of words for different objects / subjects / body parts:

"Shark": Activity in vision area because you seen one

like pick Kuch

• Self-Organizing feature Maps (SOMs)

→ Maps points from high-dimensional source space into a 2 to 3 -d target space

→ Topology: distance and proximity relationship preserved as much as possible

→ Clustering is performed by competing for the current object

→ Unit whose weight vector is closest to current object wins

→ Adjust winner and its neighbors

→ SOMs to resemble self-organization in the brain

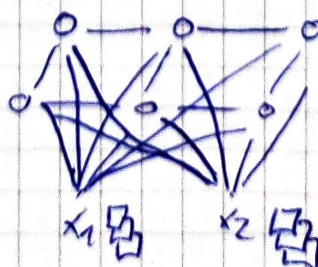
→ Kohonen map:

→ m -dimensional grid of units u (eg: $m = 2 \rightarrow u_{ij}$)

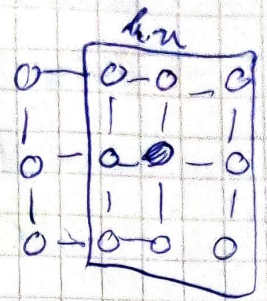
→ each unit has n -dimensional weight vector w

→ Dimensional reduction: $n \gg m$

→ Training stimuli: set of k n -dimensional vectors $X_k = \{x_{k,1}, \dots, x_{k,n}\}$



SOMs



Learning Algorithm:

- Initialisation: random weights vectors
- Present random sample x to the network
- Iterate through grid and compute distance

$$d_{ij} = \|x - w_{ij}^t\|$$

- Select best matching unit (BMU) from map, s.t. the distance is minimal

$$\arg \min_{i,j} d_{ij}$$

- Compute neighborhood h_u of BMU u

- Shift weights of units in h_u towards x → neighbourhood function

$$w_{ij}^{t+1} \leftarrow w_{ij}^t + h_u(t) \eta(t) (x - w_{ij}^t)$$

- ⇒ Over time, the network self-organises, ^{learning rate} while preserving input topology

Adaptation: $\eta(t)$

- Visualization as a distorted Kohonen map

↳ Dist between u 's $\sim$ dist in higher dimension

- Neurons are moved closer to the input pattern in high-d space

- controlled via a learning parameter $\eta(t)$ which can decay over time

Activation spreading: $h_u(t)$

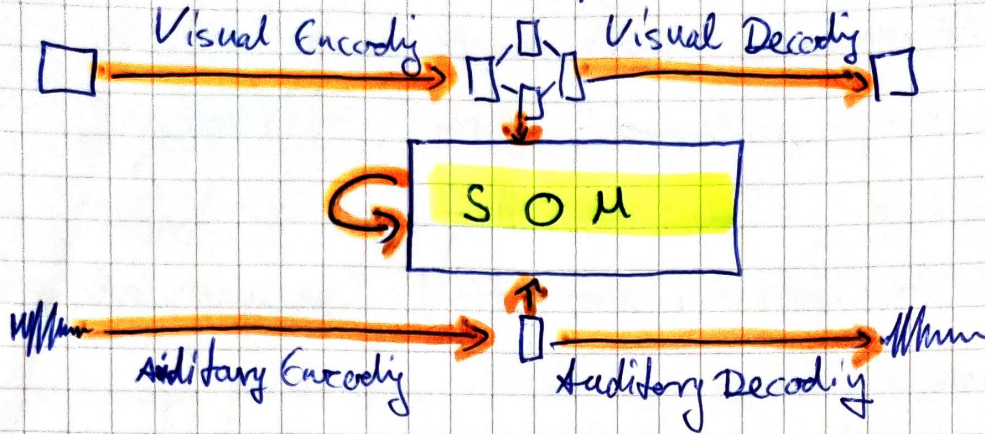
- activation of the neuron is spread in its direct neighbourhood

- neighbours become sensitive to the same input patterns (Hebb's rule)

- Size of the neighbourhood initially large but reduced over time

- How can we ground embodied sensory stimuli in embodied concepts and therefore, language?

→ Cross-modal stimuli prediction



→ Neighbourhoods form ca
→ Concepts activate latent sensory representation

- From basic to higher level grounding in supervised imitation networks

→ Epigenetic autonomous robot learns basic actions and names

→ reflect semantic associations at sensorimotor level

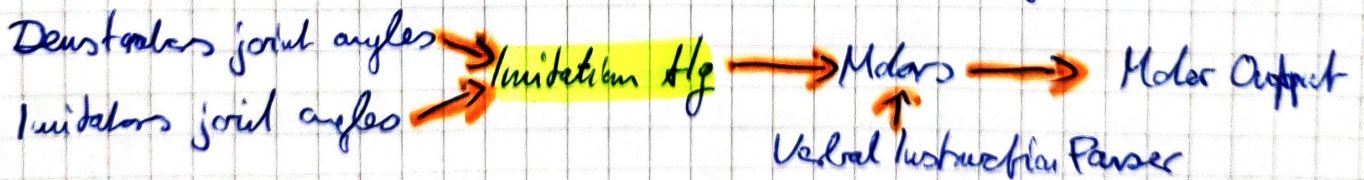
→ linguistic abilities strictly depend on motor skills and behaviours

→ Transfer learning

→ Modelled with stick-figure robots based on NN controller

→ Demonstrator and Imitator

→ Supervised neural controller for grounding



↳ Dense feed-forward NN

↳ Four pairs of motor neurons

↳ Imitation Algorithm:

→ Estimate necessary force of each motor joint

→ Minimize difference for motor output

• Learning: Basic Grounding

→ imitator learns to execute actions

Action words:

→ same start position

Open - left - from

→ observing the demonstrator

→ mimicking movement

→ learns action and names simultaneously

→ Grounding of words in perception and imitator's production

• Transfer: Higher order grounding

→ Imitator learns behaviours: combined actions, linguistic description in natural language

→ Grounding of names and concepts of new actions

• Results

Language representation allow to compose complex linguistic constructions from primitives

⇒ compositional language structure

Grounding:

Result not just accurate also contextually correct

L4

Computational models for application: Language representation with Word2Vec

→ classical representation of words in NLP:

→ Atomic words

→ Need additional taxonomy to define similarity or difference

cat [1, 0, 0]

swal [0, 1, 0]

colie [0, 0, 1]

→ Word2Vec Models:

→ Word-embedding in vector space

→ Distributed representation inherently compositional

→ Connectionist model: Simple Single-Layer Perceptron

→ Learn vectors from context or neighboring words

cat [0, 63, 0.12, 0]

Word2Vec variants

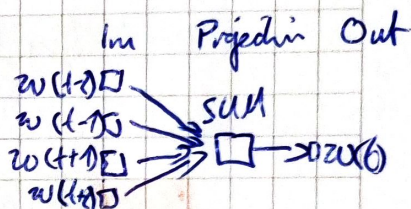
→ CBOW (Continuous Bag of Word)

→ Using context words, we can predict center word

→ Word presented as distributed probability vector

→ Captures "syntactic" relationships

→ Max: $\frac{1}{T} \sum_{t=1}^T \log p(w_t | w_{t-\frac{c}{2}}, \dots, w_{t-1}, w_{t+1}, \dots, w_{t+\frac{c}{2}})$



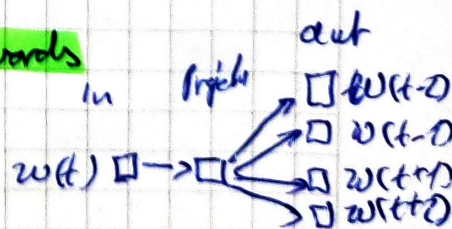
→ Skip-gram

→ With center word, we can predict context words

→ Mirror of CBOW

→ Captures "semantic" relationships

→ Max: $\frac{1}{T} \sum_{t=1}^T \sum_{-\frac{c}{2} \leq j \leq \frac{c}{2}, j \neq 0} \log p(w_{t+j} | w_t)$

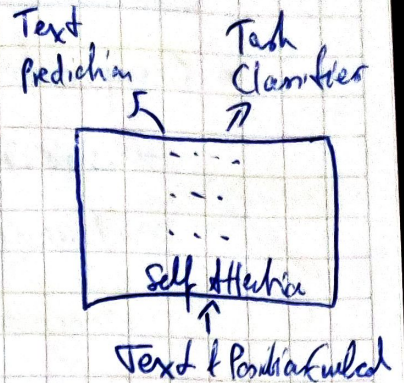


eg: Queen = King + Woman - Man
Word clustering

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• Natural Language Generation with GPT

- Based on Transformer architecture
 - Only simplified Decoder part retained
 - Stack on top of each other
- Attention instead of recurrence
- Generative pre-training:
 - guess the ^{next} word, given context
 - unsupervised
 - auto-regressive
- Enormous amount of data



• HuBERT: Self-Supervised Speech Representation

- Based on BERT
- Learning by listening:
 - learn structure of spoken input with offline clustering
 - alternate between clustering & prediction steps to progressively improve representation
- self-supervised: no reliance on supervised labels, text, etc.
- Pre-trained model can be adapted to new domains

Bio - inspired

AI

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Compositionality of Language

~~words~~

- Humans can infer alternative usages of words and understand unseen usages and new compositions of words

Guiding Training for Compositionality

- Evaluation shows that the model can infer unobserved compositions similarly to humans and is even likely to make similar mistakes

Summary

- Models for explaining human NLP

- Sensor to actuator mapping

- Semantic association

- Grounding language in action

- language acquisition / knowledge acq.

- Hebbian learning, SOM learning, Statistical learning

- Conceptual representations for stimulus prediction

- Classic NLP improved by brain-inspired representations